MILITARY SPECIFICATION SHEET
SERVOMOTOR, TYPE 18SM4a

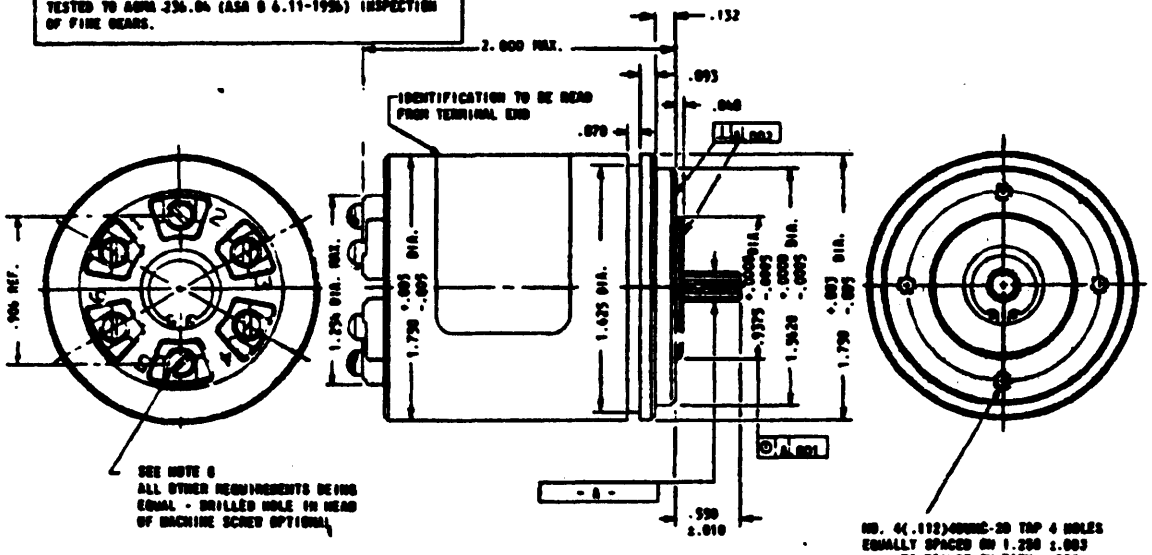
This Specification is Mandatory for Use by All Departments and Agencies of the Department of Defense.

THE COMPLETE REQUIREMENTS FOR PROCURING THE SERVOMOTOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-S-22432.

SPUR GEAR DATA	
TEETH	21
PITCH	120
OUTSIDE DIAMETER	.1972 $^{+.0000}_{-.0010}$
PRESSURE ANGLE	20°
TOOTH FORM	FULL DEPTH INVOLUTE

DESIGN TO AGMA 207.04 (ASA B 6.7-1956) "20 DEGREE INVOLUTE FINE PITCH SYSTEM" PRECISION GRADE 1
TESTED TO AGMA 236.04 (ASA B 6.11-1956) INSPECTION OF FINE GEARS.

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCE ON FRACTIONS ±1/64
DECIMALS ±.005 ANGLES ±2°



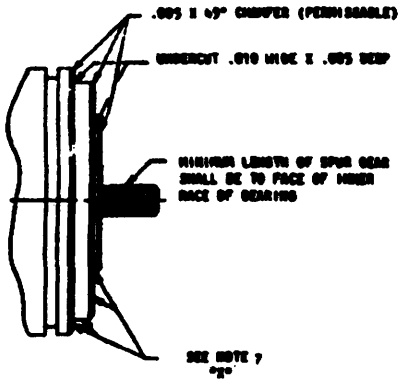


FIGURE 1

NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING SHALL BE ROTATED.
2. RUNOUT OF FEATURE "A" SHALL NOT EXCEED .0014 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1 POUND RADIAL PLAY - 1/2 POUND
4. THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS ARE REMOVED.
5. THE END CAP DESIGN SHALL BE SUCH THAT THE TERMINAL LUG MAY BE INSTALLED WITH THE BARREL INBOARD OR 180° POSITION OUTBOARD. RECESSES FOR LUGS SHALL BE OF WIDTH AND DEPTH TO PREVENT TURNING OF THE LUGS.
6. THE FOLLOWING LIST OF HARDWARE, MS-35270-313 MACHINE SCREW (8 REQUIRED), MS-35338-136 LOCK WASHER (8 REQUIRED) SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PD VALUE OF 0-0 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS, THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SERVOMOTOR PACKAGE.
7. HOUSING SHALL BE STAINLESS STEEL. SURFACES MARKED "B" SHALL BE BRIGHT METALLIC FINISH (AS PER MIL-C-13924).
8. DIMENSIONS AND TOLERANCES COMPLY WITH ANSI Y14.5.
9. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL), A SAMPLE OF THE SERVOMOTORS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN SPECIFICATION MIL-S-22432.

FSC 6105

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TABLE I (Continued)

Requirements	Value	Unit	Tolerance
Rotor Moment of Inertia	4.0	gm-cm^2	maximum
Radial Play	.0008	inch	maximum
End Play	.0045	inch	maximum
	.0020	inch	minimum
Temperature Rise	100	degrees C	maximum
Noise, Mechanical	not excessive		
Time Constant	.009	seconds	nominal
Reversing Time	.015	seconds	nominal
Theoretical Acceleration at Stall	71000	radians/second^2	nominal
Maximum Power Output	3.3	watts	nominal
Weight	12	ounces	maximum

Notes:

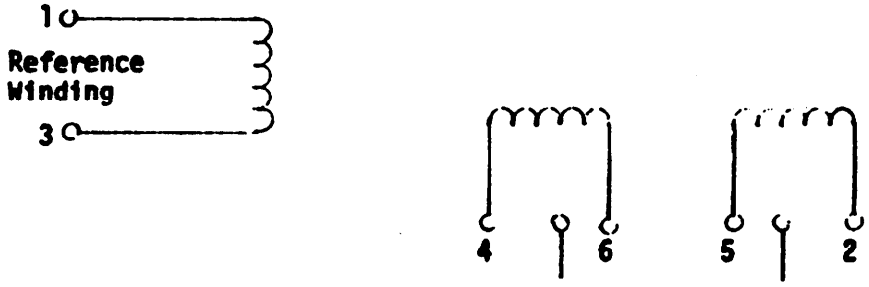
1. Type 18SM4a servomotors shall be capable of operation in ambient temperatures ranging from -55°C to 125°C.

2. The servomotor identified herein shall conform to all requirements, and shall be subjected to all tests delineated in MIL-S-22432, with the following exceptions:

Salt Atmosphere Resistance, paragraph 3.30
Audible Noise, paragraph 3.31

3. The type 18SM4a servomotor was formerly identified as the BUORD Mark 8 Mod 3.

4. Connection Diagram:



Control Winding Excitation 400 cps

Figure 2

a. For CCW rotation the voltage at terminal 2 leads the voltage at terminal 1 by 90 electrical degrees.
b. For series operation of control winding terminals 5 and 6 are strapped and 115 volts rated is applied across terminals 2 and 4.
c. For multiple or parallel operation of control winding terminals 6 and 2 are strapped, and terminals 4 and 5 are strapped. 57.5 volts rated is applied across terminals 2 and 4.

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OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-22432/11B SERVOMOTOR, TYPE 18SM4a

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS.

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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